Similarly, the flow map of the system with Hamiltonian $H_2 = U$ is

$$\begin{aligned} \boldsymbol{Q} &= \boldsymbol{q}, \\ \boldsymbol{P} &= \boldsymbol{p} - h\nabla U(\boldsymbol{q}). \end{aligned}$$

The composition of these maps simplifies to

$$\begin{aligned} \boldsymbol{Q} &= \boldsymbol{q} + h\boldsymbol{M}^{-1}\boldsymbol{P}, \\ \boldsymbol{P} &= \boldsymbol{p} - h\nabla U(\boldsymbol{q}), \end{aligned}$$

which is precisely the Symplectic Euler method. Similarly, composing the same two maps in the opposite order gives the adjoint Symplectic Euler method.

Example 2.5 (Verlet is Symplectic) For the Symplectic Euler method $\mathscr{G}_h$ and its adjoint method $\mathscr{G}_h^\dagger$, consider the composition

$$\mathcal{K}_h := \mathscr{G}_{h/2}^\dagger \circ \mathscr{G}_{h/2}.$$

Let

$$\begin{bmatrix} \bar{\boldsymbol{q}} \\ \bar{\boldsymbol{p}} \end{bmatrix} = \mathscr{G}_{h/2} \begin{bmatrix} \boldsymbol{q} \\ \boldsymbol{p} \end{bmatrix},$$

i.e.,

$$\bar{\boldsymbol{q}} = \boldsymbol{q} + (h/2)\boldsymbol{M}^{-1}\bar{\boldsymbol{p}}, \qquad \bar{\boldsymbol{p}} = \boldsymbol{p} - (h/2)\nabla U(\boldsymbol{q}).$$

and set

$$\begin{bmatrix} \boldsymbol{Q} \\ \boldsymbol{P} \end{bmatrix} = \mathscr{G}_{h/2}^\dagger \begin{bmatrix} \bar{\boldsymbol{q}} \\ \bar{\boldsymbol{p}} \end{bmatrix},$$

i.e.,

$$\boldsymbol{Q} = \bar{\boldsymbol{q}} + (h/2)\boldsymbol{M}^{-1}\bar{\boldsymbol{p}}, \qquad \boldsymbol{P} = \bar{\boldsymbol{p}} - (h/2)\nabla U(\boldsymbol{Q}).$$

The composition simplifies to

$$\begin{aligned} \bar{\boldsymbol{p}} &= \boldsymbol{p} - \frac{h}{2}\nabla U(\boldsymbol{q}), \\ \boldsymbol{Q} &= \boldsymbol{q} + h\boldsymbol{M}^{-1}\bar{\boldsymbol{p}}, \\ \boldsymbol{P} &= \bar{\boldsymbol{p}} - \frac{h}{2}\nabla U(\boldsymbol{Q}). \end{aligned}$$

This is just the leapfrog/Verlet method in its Hamiltonian form. Since we have obtained the method as the composition of two symplectic maps, and the symplectic maps form a group, we know that this method will also be symplectic.

When we construct a composition of a method with its adjoint method, we see readily that it is symmetric, i.e. self-adjoint, since

$$\mathcal{K}_h^\dagger := [\mathcal{G}_{h/2}^\dagger \circ \mathcal{G}_{h/2}]^\dagger = \mathcal{G}_{h/2}^\dagger \circ \mathcal{G}_{h/2} = \mathcal{K}_h.$$

Self-adjoint, or *symmetric* schemes have unique features; in particular they have even order [164, 227].

As we shall see in the next chapter, it is possible to construct methods of arbitrary order by employing more sophisticated multi-stage compositions of mappings.

2.4.2 General Composition Methods

When using splittings, it is not necessary to solve each Hamiltonian of a splitting using the exact flow. Instead, we may replace the flow maps of any part by an approximation. More generally, if we have any two symplectic numerical methods, say $\mathcal{G}_h^1$ and $\mathcal{G}_h^2$, then the composition

$$\mathcal{G}_h := \mathcal{G}_{h/2}^1 \circ \mathcal{G}_{h/2}^2$$

is another symplectic numerical method. The order of this method is typically the minimum of the orders of the two methods involved, but it can be higher, as the example of the Verlet method (constructed by composing symplectic Euler and its adjoint) shows.

2.4.3 Harmonic + Anharmonic Splitting

Some systems can be decomposed into a harmonic (quadratic) part $H_0(\boldsymbol{q},\boldsymbol{p}) = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2 + \boldsymbol{q}^T\boldsymbol{A}\boldsymbol{q}/2$ and an anharmonic part $H_1(\boldsymbol{q},\boldsymbol{p}) = \hat{U}(\boldsymbol{q})$. Then a splitting method can be formulated based on exact solution of the linear system together with "kicks" representing impulses defined by the anharmonic part. For a simple system consisting of a harmonic oscillator $H_0(q,p) = p^2/2 + \Omega^2 q^2/2$, such a method can be written

$$\begin{bmatrix} Q \\ \hat{p} \end{bmatrix} = \begin{bmatrix} \cos h\Omega & \frac{1}{\Omega}\sin h\Omega \\ -\Omega \sin h\Omega & \cos h\Omega \end{bmatrix} \begin{bmatrix} q \\ p \end{bmatrix},$$
$$P = \hat{p} - h\hat{U}'(Q).$$